MTH 202 Chapter 3: Partial Derivatives

3.1 Functions of Two or More Variables

Definition: A <u>function f of two variables</u> is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the <u>domain</u> of f and its <u>range</u> is the set of values that f takes on, that is, $\{f(x, y) \mid (x, y) \in D\}$.

Note: Familiar operations like addition can be written as a two variable function. For example, the function $f(x, y) = x + y$ is the function which spits out the sum of two numbers. Using our mapping notation we can write $f : \mathbb{R}^2 \to \mathbb{R}$.

<u>Example 1:</u> Consider $f(x, y) = \dfrac{\sqrt{x+y+1}}{x-1}$.

a) Find $f(2,1)$.

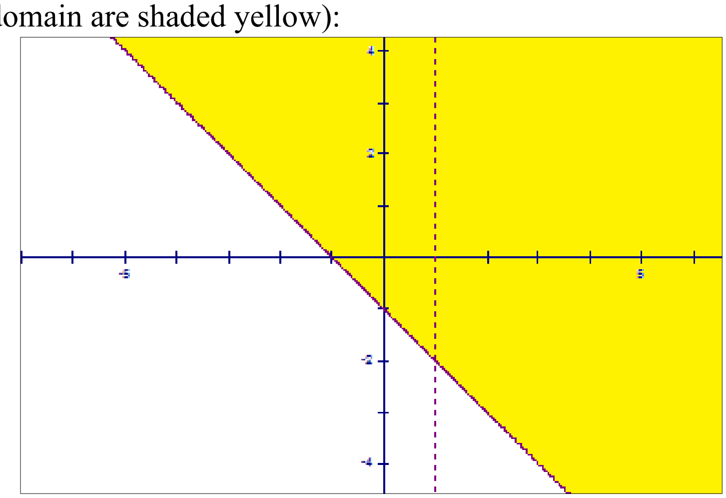
We note $f(2,1) = \dfrac{\sqrt{2+1+1}}{2-1} = 2$.

b) Write the domain of $f(x, y)$ in set notation.

In order for (x, y) to be in the domain of f, we need $x + y + 1 \geq 0$ and $x - 1 \neq 0$. Thus the domain of f is $D = \{(x, y) \mid x + y + 1 \geq 0,\ x \neq 1\}$.

c) Sketch a picture of the domain of f.

A picture is (points in the domain are shaded yellow):



Example 2: Write the domain and range of the following function in set notation: $f(x,y)=\sqrt{4-x^2-y^2}$.

First, for (x,y) to be in the domain we need $4-x^2-y^2 \geq 0 \Leftrightarrow x^2+y^2 \leq 4$. Thus, the domain of the function is $\{(x,y) \mid x^2+y^2 \leq 4\}$. Since x^2 and y^2 are both nonnegative numbers, we know that $f(x,y)$ can be no larger than $\sqrt{4}=2$, and in fact, $f(0,0)=2$. Also, since the range is not to include complex numbers, the smallest value that $f(x,y)$ can be is 0, and in fact, $f(2,0)=0$. So the range of the function is $\{z \mid 0 \leq z \leq 2\}$.

Exploration:
From the above example we see that we can think of $f(x,y)$ as the *z* variable. Thus, we can graph a function of two variables in $\mathbb{R}^3$. Moreover, the graph of a two variable function is a surface in $\mathbb{R}^3$.

The instructor will use the three dimensional plotter to graph the function from the most recent example. The instructor will also graph it by hand (the graph is a hemisphere).

Intuitively, when we graph a two variable function we can think of plotting points, above, below, or in the domain region (there are nice illustrations of this throughout section 13.1 in the book).

Definition: The <u>level curves</u> of a function f of two variables are curves with equations $f(x,y)=k$, where *k* is a constant in the range of f.

Note: Level curves are closely related to trace curves, and are very helpful when sketching a graph of a two variable function by hand.

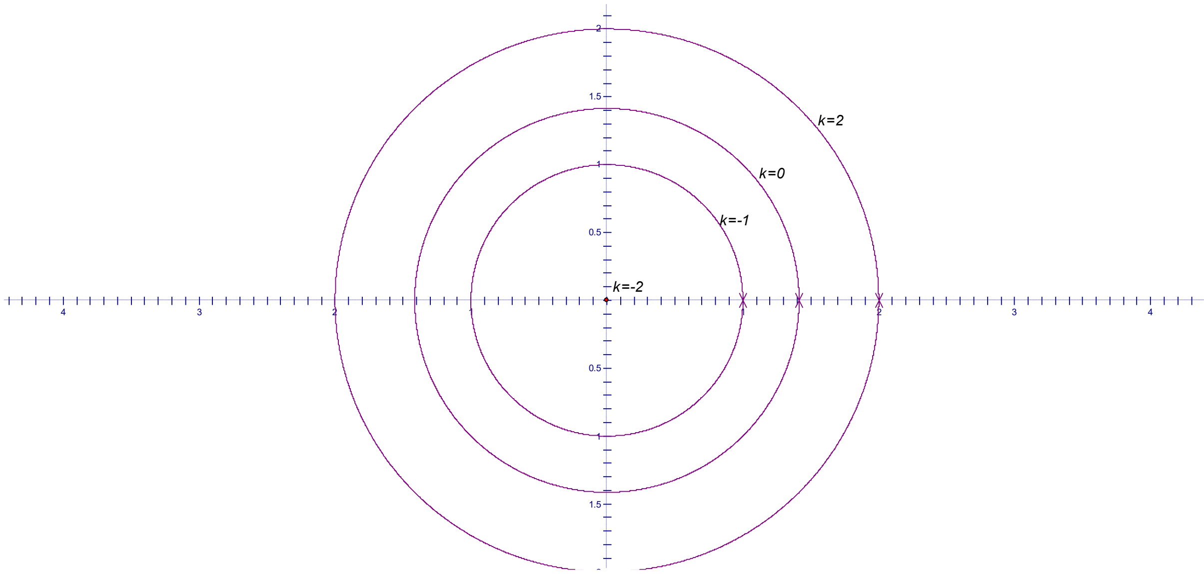
Example 3: Consider the function $f(x,y)=x^2+y^2-2$.

a) State the domain and range of this function.

Since plugging any (x,y) in f will produce a real number output, we know that the domain is $\mathbb{R}^2$. The smallest that f can be is $f(0,0)=-2$, and f can get arbitrarily large. So, the range of the function is $\{z : -2 \leq z\}$.

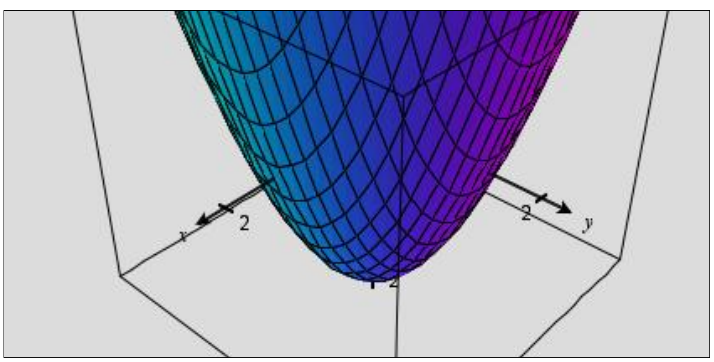
b) Sketch the level curves for the function for the values $k = -2, -1, 0, 2$.

We note that if $f(x,y) = k$, then $x^2 + y^2 = k + 2$. So, the level curves are:



c) Sketch a graph of $f(x,y)$.

A sketch is:



Note: We recognize the function in this example as having a graph which is an elliptic paraboloid.

Note: Two dimensional graphs of level curves (as we saw in part (b)) of the most recent example are used frequently in topographic maps of mountainous regions and are called <u>contour maps</u>.

Question to the Class: What does the graph of $f(x,y) = x + y - 2$ look like?

This linear function represents a plane with normal vector $\langle -1, -1, 1 \rangle$.

Definition: A <u>function of *n* variables</u> is a rule that assigns a number $z = f(x_1, x_2, \ldots, x_n)$ to an *n*-tuple $(x_1, x_2, \ldots x_n) \in \mathbb{R}^n$.

Remark: If $n \geq 3$ functions of *n* variables can be very difficult or impossible to picture visually. One simple example of how one might picture a function of 3 variables is representing how a surface changes over time.

For example, consider the function $f(x, y, z) = x^2 + y^2 + z^2$. If we think of $f(x, y, z)$ as being a time variable, the surface is a sphere of radius one when time is one, a sphere of radius 2 when time is 4, etc (*Note: This is certainly not the only way one might interpret this function*).

<u>**Example 4:**</u> At a bakery there are *n* ingredients used to make all possible items. Suppose that x_i represents the number of units of the i^{th} ingredient used by the bakery on a particular day. Suppose also that the i^{th} ingredient costs c_i per unit.

a) If $C(x_1, x_2, \ldots, x_n)$ is the daily expense function (on ingredients) for the bakery, find a formula for $C(x_1, x_2, \ldots, x_n)$.

   One way to write it would be $C(x_1, x_2, \ldots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n$. Another way to write it would be $C(x_1, x_2, \ldots, x_n) = \sum_{i=1}^{n} c_i x_i$.

b) Write the domain of the function in (a) in set notation.

   We assume that the bakery cannot use a negative amount of an ingredient. So, the domain is $\{(x_1, x_2, \ldots, x_n) \mid x_1 \geq 0,\ x_2 \geq 0, \ldots, x_n \geq 0\}$.